Naramamidi Raja Keerthana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.16
SRI SHIRIDI SAI VIDYANIKETAN E M SCHOOL,KADIAM,ANDHRA PRADESH	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2023	91.20%
DR KKRS GOWTHAM HIGH SCHOOL HUKUMPETA ,EAST GODAVRI DISTRICT,ANDHRA PRADESH	Class X (BOARD OF SECONDARY EDUCATION , ANDHRA PRADESH)	2021	98.30%

WORK EXPERIENCE

AI Engineer Intern

- Developed and tested **API integrations and Microsoft connector components** for Outlook Mail, Excel, Microsoft To Do, and other Microsoft services.
- Designed and mapped **AI agent workflows** for Sales, Customer Service, and Inventory use cases across Executive, Manager, and Director levels, identifying opportunities for **agent-based automation** in retail operations.
- Built and tested **coding-agent and LLM-based workflows**, integrating APIs, tools, and business data to support automated task execution and enterprise use cases.
- Used **OpenAI Codex for AI-assisted coding**, debugging, and resolving integration issues across connector and workflow components.
- Worked with **Large Language Models (LLMs), API integrations, agent workflows, Microsoft Graph, Docker, Git, and Python** during development and testing.

PROJECTS

Customer Churn Intelligence Dashboard | Python, XGBoost, SHAP, Streamlit

- Built an end-to-end **customer churn prediction pipeline** comparing Logistic Regression, Random Forest, and XGBoost, with **SMOTE-based class balancing and hyperparameter tuning**.
- Applied **SHAP** for global feature importance and customer-level prediction explanations, identifying key **churn drivers and high-risk customer segments**.
- Developed a **5-page Streamlit decision dashboard** featuring executive KPIs, batch prediction with downloadable results, customer-level analysis, what-if scenario analysis, and **model explainability**.

MOOC Product Funnel & Learner Engagement Analytics | SQL, Python, Power BI

- Analyzed **338K+ learner-course records** using **Python and SQL (DuckDB)** to build a registration ->view ->explore -> certification funnel and calculate **stage-wise conversion rates**.
- Developed an interactive **Power BI dashboard** with course, engagement, education, and gender segmentation to analyze **learner behavior and funnel conversion patterns**.
- Segmented learners by recorded event activity using SQL and identified a **10.81% observed certification rate** among learners with **50+ events**, compared with **0.17%** for learners with **10-49 events**.

SKILLS

Programming Languages: Python, SQL,Bash

Libraries & Tools: Pandas, NumPy, PyTorch, TensorFlow, Matplotlib, Plotly, Power BI, Excel,Streamlit, DuckDB, Git, GitHub, Docker, Jupyter, VS Code

AI / GenAI: RAG, LLMs, Prompt Engineering, Agent Workflows, FAISS, BM25, Cross-Encoder, Gemini

Machine Learning: Supervised Learning, Classification, Regression, Random Forest, XGBoost, Logistic Regression, SVM, KNN, Model Evaluation, SHAP

Soft Skills: Problem Solving, Analytical Thinking, Communication, Teamwork, Adaptability

RELEVANT COURSES

Mathematics: Probability and Random Variables, Elementary Linear Algebra, Biostatistics

Programming/AI: Introduction to Programming, Artificial Intelligence, Machine Learning in Bioinformatics, R and Python for Biologists,AI in Drug Discovery

Data: Business Analytics using Python, Data Science Analysis

SCHOLASTIC ACHIEVEMENTS

- Secured **3rd class rank and 187th Olympiad rank** in **International Master Mathematics Olympiad** 2015.

EXTRACURRICULAR

NSS Volunteer, IIT Hyderabad

Participated in walkathons, Swachh Bharat and Clean India drives, Vidyadan initiatives, and community outreach activities.